Harshil Modh

Hoboken, NJ, USA | (551) 267-0683 | harshilmodh77@gmail.com | LinkedIn | GitHub | Portfolio

Education

Stevens Institute of Technology | Master of Science in Computer Science; GPA: 3.86/4.00 — 2024 - May 2026

LDRP Institute of Technology and Research | Bachelor of Engineering, Computer Science — 2019 - 2023

Achievement: Winner, AWS Publishing & Media Innovation Hackathon, NYC (CoMP Ledger) — July 2026

Experience

Software Engineer Intern, AI — Jan 2026 - May 2026

GenAI Consulting | *React 19, TypeScript, Tailwind CSS, Python, FastAPI, PostgreSQL, OpenAI API, Azure, pytest* — *Princeton, NJ*

Built a full-stack AI sales platform for a pharmaceutical client's field team, serving live queries over 71K+ rows across 10 tables.

- Built the React/TypeScript rep app that tells reps which doctors to visit next and briefs them before each visit.
- Engineered the FastAPI data layer with startup dimension-table caching and filtered SQL over 41K TRx and 27K call records.
- Built a RAG assistant that refuses to answer unverifiable numbers; answered about 96% of 70 live test questions in 2-4 seconds.
- Added fail-closed guardrails in Python so the AI only drafts text and blocks patient data and off-label requests.
- Deployed on Azure Container Apps with managed identity (no stored credentials) and scale-to-zero, removing $263/month idle cost.
- Backed the platform with 760+ pytest tests and hand-reviewed every change before merge.

Teaching Assistant, CS 546 Web Programming — Sept 2025 - Aug 2026

Stevens Institute of Technology | *Node.js, Express, MongoDB, npm, Canvas API, Git* — *Hoboken, NJ*

- Built and published canvas-grade-manager, an npm CLI on the Canvas API that automates late penalties, bulk submission downloads, and grading reports; 1,300+ weekly downloads, adopted by 6+ TAs over two semesters.
- Built a grading framework that checks HTTP responses, JSON contracts, auth, and database state, cutting grading turnaround 80%.
- Debugged 50+ student defects in async race conditions, middleware pipelines, and auth flows, cutting resolution time 60%.
- Mentored 100+ graduate students through 80+ weekly code reviews, cutting critical defects 30%.

Software Engineer — June 2023 - June 2024

Grownited | *Node.js, Express, MongoDB, Redis, React, Angular, Docker, AWS, GitHub Actions* — *India*

- Architected modular CRM backends in Node.js, Express, and MongoDB with layered services and standardized API contracts; secured 15+ production endpoints using JWT, RBAC, validation middleware, and authorization failure-path tests.
- Cut API latency 50% and database reads 40% through query-plan analysis, aggregation and index tuning, and Redis TTL caching.
- Built role-aware React and Angular dashboards with protected navigation, multi-step form validation, and reusable Google Maps search components, cutting reporting turnaround 35%.
- Containerized services with Docker and shipped to AWS through GitHub Actions CI/CD; reviewed PRs and mentored juniors.

Projects

CodePulse, AI Code Health Platform | GitHub *Node.js, Express, PostgreSQL, Prisma, BullMQ, Redis, pgvector, OpenAI, Next.js*

- Architected a Node.js/Express pipeline that scores a GitHub repo 0-100 within 60s of each push, running five analyses in parallel through BullMQ fan-out/fan-in: complexity, vulnerabilities, dead code, coverage, and drift.
- Built OpenAI agents for six-tool RAG code chat, root-cause investigation, multi-agent design debate, and codebase tours; streamed over SSE with sub-400ms time-to-first-response and exposed the agents to IDEs through an MCP server.
- Detected semantic architectural drift with OpenAI embeddings and pgvector cosine search in under 50ms; grouped related files with K-Means on a multi-tenant PostgreSQL/Prisma schema of 14 models.
- Built the Next.js/TypeScript client with Clerk auth, Stripe billing tiers, HMAC-verified webhooks, and hashed API keys; pushed PR health comments and live score updates through Redis Pub/Sub and Socket.IO.

CareConnect, Real-Time Caregiving Platform | GitHub | Live *React, Node.js, Express, MongoDB, Redis, Socket.IO, AWS*

- Built a React, Node.js, Express, and MongoDB platform for caregiving tasks, medications, vitals, and documents; secured 40+ REST endpoints with five-role RBAC, per-membership permission checks, and JWT plus Firebase authentication.
- Reduced database queries 50% using Redis caching and token-based stateless sessions; deployed a four-service Docker stack on AWS EC2 and secured document access with S3 pre-signed URLs.
- Engineered Socket.IO group chat with edit history, read receipts, and admin-only deletion; added one-tap emergency alerts reaching all group members through in-app notifications, chat, and email.
- Automated four background jobs for recurring tasks, overdue detection, deduplicated reminders, and notification cleanup; kept medication dose tracking accurate under concurrent caregiver writes.

CoMP Ledger, AI Content Attribution Platform | GitHub *Node.js, Express, PostgreSQL, pgvector, AWS Bedrock, S3*

- Built the AWS hackathon-winning platform that tests whether an AI answer drew on a publisher's reporting: a Node.js, Express, and PostgreSQL backend with 40 REST endpoints, JWT auth, and rate limiting.
- Engineered RSS-driven corpus ingestion and a verification pipeline that checks every claim against five external sources (Wikipedia, Google Search, OpenAlex, GDELT, Wayback Machine) before a Claude judge returns a structured verdict.
- Designed the exclusivity scoring model, scoring each claim inversely to how many independent sources carry it, so common knowledge scores low and original reporting scores high.
- Streamed scored claims to the React client over Server-Sent Events; built tamper-evident SHA-256 signed exports to S3, with the hash mirrored in object metadata, for licensing negotiations.

Skills

Languages: JavaScript, TypeScript, Python, SQL, Java
Backend: Node.js, Express, FastAPI, Flask, REST APIs, GraphQL, Socket.IO, SSE, Redis, BullMQ, Zod
Frontend: React, Next.js, Angular, Vite, Tailwind CSS, shadcn/ui, TanStack Query, D3, Recharts
Databases: PostgreSQL, MongoDB, pgvector, MySQL, Firestore, Prisma
Cloud / DevOps: AWS EC2, AWS S3, AWS Bedrock, Azure Container Apps, Docker, GitHub Actions, CI/CD, Vercel, Linux
AI / LLM: OpenAI API, Claude, Gemini API, Amazon Titan, RAG, MCP, Embeddings, Claude Code
Auth / Testing: JWT, OAuth, RBAC, HMAC-SHA256, Rate Limiting, Clerk, Firebase Auth, Stripe, pytest, Postman, Git